

HOSPITAL APPOINTMENT BOOKING SYSTEM

1. Title

Hospital Appointment Booking System using HTML, CSS, JavaScript, Node.js, and PostgreSQL

2. Abstract

The Hospital Appointment Booking System is a web-based application designed to simplify and automate the process of booking appointments between patients and doctors. It allows patients to view available doctors, check schedules, and book appointments online without visiting the hospital physically. This system reduces waiting time, improves efficiency, and provides a better user experience for patients.

The system also enables administrators to manage doctor details, appointment schedules, and patient records efficiently. It ensures proper organization of hospital operations and minimizes manual errors. The application is developed using HTML, CSS, JavaScript, and Bootstrap for frontend design, Node.js for backend processing, and PostgreSQL for database management, making it reliable and scalable.

3. Introduction

In traditional hospitals, appointment booking is often done manually, leading to long waiting times and inefficiencies. Patients must visit hospitals physically or call to book appointments, which is time-consuming.

The Hospital Appointment Booking System provides a digital platform where patients can easily schedule appointments online. It improves communication between patients and hospital staff and ensures better management of time and resources.

4. Problem Statement

The existing system faces several challenges:

- Manual appointment booking
- Long waiting times
- Difficulty in managing doctor schedules
- Chances of overlapping appointments
- Lack of centralized system

5. Aim of the Project

The aim of this project is to develop a web-based system that:

- Enables online appointment booking
 - Reduces waiting time
 - Manages doctor schedules efficiently
 - Maintains patient records
 - Improves hospital management
-

6. Existing System

- Appointment booking is manual
 - No proper schedule management
 - Time-consuming process
 - High chances of errors
 - No digital records
-

7. Proposed System

The proposed system is a web-based application that:

- Allows patients to book appointments online
 - Displays doctor availability
 - Stores data in PostgreSQL database
 - Provides admin control for managing doctors and appointments
 - Ensures efficient scheduling and record management
-

Modules of the Project

Module 1: Authentication Module

- Patient Registration
- Patient Login
- Admin Login
- Secure access

Module 2: Admin Module

(Admin controls the system)

Operations:

- Add doctor details
 - Update doctor schedules
 - Delete doctor records
 - View all appointments
 - Manage patient records
-

Module 3: Patient Module

- Register and login
 - View available doctors
 - Book appointments
 - View appointment history
-

Module 4: Doctor Management Module

- Add doctor details
 - Manage specialization
 - Set availability schedule
-

Module 5: Appointment Management Module

- Book appointments
 - Cancel appointments
 - Track appointment status
 - Manage scheduling
-

Module 6: Reports Module

- View appointment records
- Generate reports
- Analyze hospital activity

9. Technologies Used and Purpose

Technology	Purpose
HTML	Structure of web pages
CSS	Styling and layout
JavaScript	Client-side interaction
Bootstrap	Responsive UI design
Node.js	Backend server
PostgreSQL	Database management

11. Conclusion

The Hospital Appointment Booking System provides an efficient solution for managing appointments digitally. It reduces manual work, improves scheduling accuracy, and enhances patient experience. The system ensures better organization and effective hospital management.

12. Future Enhancements

The system can be enhanced by adding:

- Online consultation feature
- SMS/Email notifications
- Online payment system
- Mobile application support
- AI-based scheduling